

Results Interpretation Report

In Silico Molecular Docking Analysis of Natural Bioactive Compounds (Wogonin, β -Sitosterol, Baicalein, and Berberine) Targeting PPARG in Atherosclerosis.

Previously, network pharmacology analysis identified peroxisome proliferator-activated receptor gamma (PPARG) as a central hub protein associated with the PPAR signaling pathway, which is relevant to the molecular mechanisms of atherosclerosis. Therefore, in this study, molecular docking was performed to investigate the potential interactions between PPARG and the selected bioactive compounds that were identified as potential targets of PPARG in the compound–target–pathway network, namely, Wogonin, β -Sitosterol, Baicalein, and Berberine.

First, the three-dimensional structure of PPARG was obtained from the RCSB Protein Data Bank (PDB ID: 7AWC), while the molecular structures of the selected bioactive compounds were obtained from PubChem in SMILES format. Then, molecular docking was performed using SwissDock, a web-based platform for protein–ligand docking. The docking analysis was conducted using the defined binding pocket of 41, 4, 84 and consistent sampling parameters for all compounds to allow comparison of their predicted binding affinities toward PPARG.

Table 1. Protein target and ligand information used for molecular docking analysis.

Protein Target		
Name	PDB ID	Resolusi
PPARG	7AWC	1.74 Å
Ligan		
Name	SMILES	
Wogonin	<chem>COC1=C(C=C(C2=C1OC(=CC2=O)C3=CC=CC=C3)O)O</chem>	
β -Sitosterol	<chem>CC[C@H](CC[C@@H](C)[C@H]1CC[C@@H]2[C@@]1(CC[C@H]3[C@H]2CC=C4[C@@]3(CC[C@@H](C4)O)C)C(C)C</chem>	
Baicalein	<chem>C1=CC=C(C=C1)C2=CC(=O)C3=C(O2)C=C(C(=C3O)O)O</chem>	
Berberine	<chem>COC1=C(C2=C[N+]3=C(C=C2C=C1)C4=CC5=C(C=C4CC3)OCO5)OC</chem>	

Model	Calculated affinity (kcal/mol)
1	-6.735
2	-6.726
3	-6.553
4	-6.402
5	-6.367
6	-6.027
7	-5.812
8	-5.545
9	-5.119
10	-4.569
11	-4.550
12	-4.310

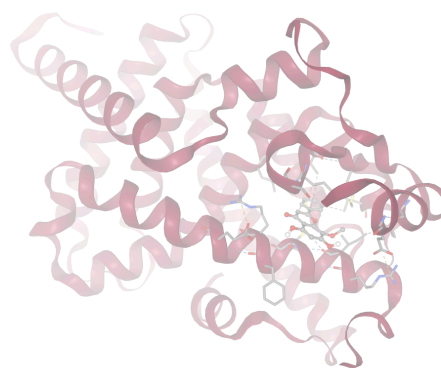


Figure 1. Binding affinity and protein–ligand interaction visualization of Wogonin with PPARG.

Model	Calculated affinity (kcal/mol)
1	-7.051
2	-6.628
3	-6.219
4	-6.001
5	-5.849
6	-4.139

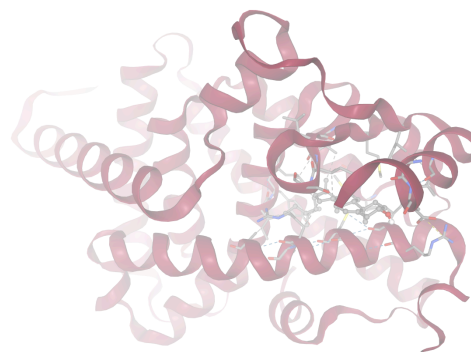


Figure 2. Binding affinity and protein–ligand interaction visualization of β -Sitosterol with PPARG.

Model	Calculated affinity (kcal/mol)
1	-7.121
2	-7.115
3	-7.094
4	-6.948
5	-6.919
6	-6.872
7	-6.799
8	-6.625
9	-6.536
10	-6.516
11	-6.021
12	-6.000
13	-5.948
14	-5.764
15	-5.647
16	-5.235
17	-5.205
18	-4.187

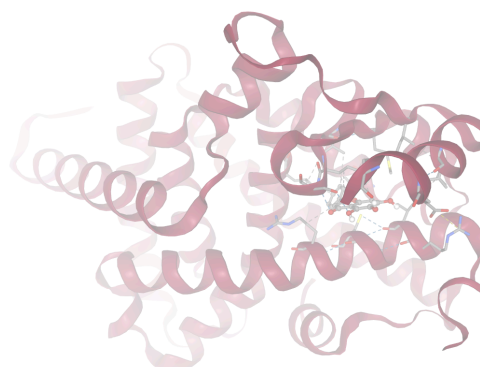


Figure 3. Binding affinity and protein–ligand interaction visualization of Baicalein with PPARG.

Model	Calculated affinity (kcal/mol)
1	-7.421
2	-7.167
3	-7.101
4	-6.969
5	-6.806
6	-6.381
7	-6.228
8	-6.130
9	-6.024
10	-4.607

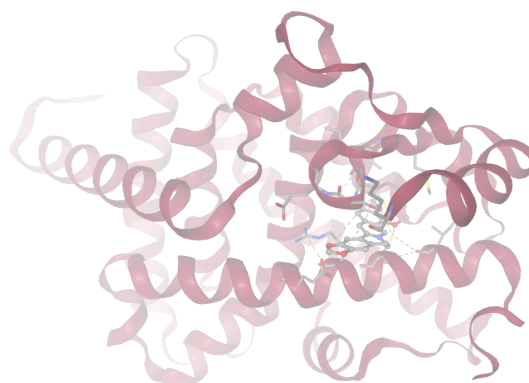


Figure 4. Binding affinity and protein–ligand interaction visualization of Berberine with PPARG.

Based on Model 1, representing the best-scoring docking pose, the results showed that Berberine had the most favorable predicted binding affinity to PPARG at -7.421 kcal/mol, followed by Baicalein -7.121 kcal/mol, β -Sitosterol -7.051 kcal/mol, and Wogonin -6.735 kcal/mol. More negative binding affinity values generally indicate more favorable predicted protein–ligand interactions. However, the stronger predicted binding affinity of Berberine does not necessarily mean that it is the most effective compound against atherosclerosis. Further analysis of protein–ligand interactions and additional computational approaches, as well as experimental validation would be required to support its potential biological activity.